

STRUCTURAL LOAD ANALYSIS OF RCC SLAB (Residential Building)

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1. Introduction

This report presents structural load analysis of an RCC slab. Dead, live, wind and earthquake loads are estimated using standard educational assumptions.

Parameter	Value
Slab Size	12 m × 10 m
Area	120 m ²
Thickness	0.15 m
Density	25 kN/m ³
Live Load	2 kN/m ²
Wind Load	1 kN/m ²
Earthquake Load	10% of Dead Load

2. Calculations

Dead Load = $120 \times 0.15 \times 25 = 450$ kN

Live Load = $120 \times 2 = 240$ kN

Wind Load = $120 \times 1 = 120$ kN

Earthquake Load = $10\% \times 450 = 45$ kN

Total Load = 855 kN

3. Discussion

Dead load is the major permanent load. Live load represents occupancy. Wind load accounts for lateral wind effects. Earthquake load is taken as 10% of the dead load for educational purposes only. The combined load is 855 kN.

4. Conclusion

The calculated total load on the slab is 855 kN. Based on these assumed values, the slab is considered satisfactory for classroom analysis. Actual structural design must comply with IS 456:2000, IS 875 (Parts 1–3), and IS 1893.